

BUDDY

Affordable AI-Powered Intelligent Robot Dog

An affordable, integrated, embodied robotic platform combining offline auditory intelligence, face & emotion perception, and autonomous obstacle-supervised mobility for interactive companionship and home awareness.

1. COMPANIONSHIP

Natural voice & visual interaction for isolated individuals

2. HOME AWARENESS

Embodied mobile patrol & safety-first sensing

3. ACCESSIBILITY

Edge-oriented processing with low-cost components



RESEARCH TEAM & CONTRIBUTIONS

- **Ashandth U.** — Offline Voice / Auditory Intelligence
- **Sobiya A. S.** — Face / Emotion Intelligence
- **SSY Wickramasinghe** — Autonomous Navigation / Patrol

EXISTING PARADIGMS VS. BUDDY GAP

Social / Companion Robots **✗ High Cost / Proprietary**
Rich social interaction but cost-prohibitive for average households.

Fixed Smart Home Monitors **✗ No Embodied Interaction**
Sensors/cameras provide alerts but lack mobility & social presence.

Cloud Voice Assistants **✗ Cloud / Privacy Dependency**
Strong NLP but immobile and transmits audio/video to external clouds.

THE IDENTIFIED RESEARCH GAP
"Limited availability of an affordable embodied robotic platform integrating privacy-aware interaction, perception and autonomous home-security functions."

THREE RESEARCH CONTRIBUTIONS → ONE BUDDY SYSTEM

Ashandth — Auditory Intelligence **EDGE VERIFIED**
Offline wake-word engine, localized STT & intent mapping without cloud transmission for strict audio privacy.

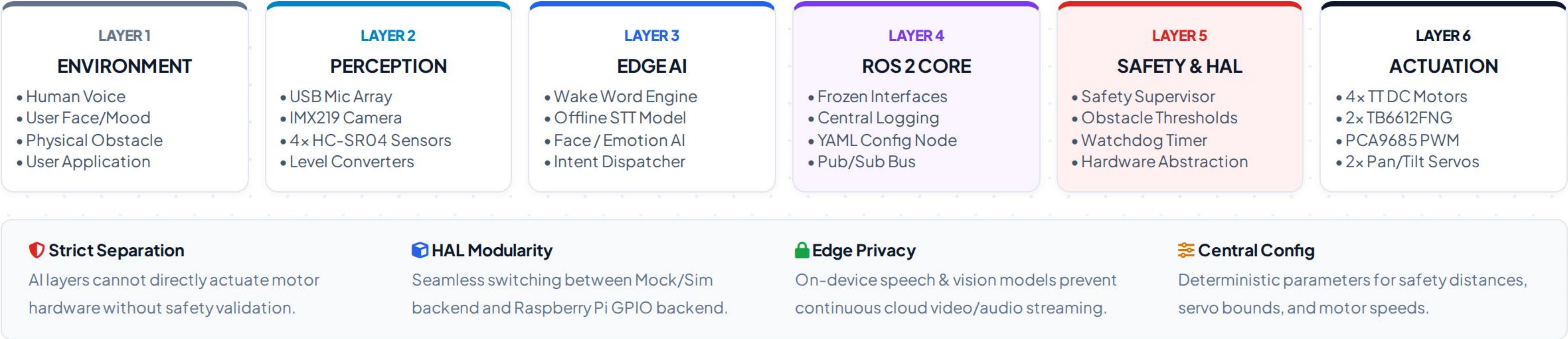
Sobiya — Face & Emotion Intelligence **ON-CHIP VALIDATION**
Edge-oriented face recognition & probabilistic emotion perception integrated with active pan/tilt head tracking.

SSY Wickramasinghe — Navigation & Safety **PHYSICALLY TESTED**
Obstacle-supervised patrol logic, dual-level distance thresholding, fail-safe motor supervision & web monitoring.

ONE ROBOT. MULTIPLE CAPABILITIES. ONE COORDINATED ROS 2 CORE.

One Integrated Layered Architecture

MARKING CAT 2 & 3 (40%)



⚠ CORE SAFETY PRINCIPLE

"AI does not directly control safety-critical motor hardware."

All movement requests from AI, user app, or patrol routines are intercepted and verified by the independent Safety Supervisor node.

CONFIGURED SAFETY THRESHOLDS (HC-SR04)

CAUTION ZONE

0.45 m

Speed reduction & alert

EMERGENCY STOP

0.25 m

Zero-velocity hardware clamp

Fail-Safe Features: Watchdog heartbeat, safe servo pan/tilt software limits (PCA9685), and hardware-disabled-by-default initialization.

END-TO-END PHYSICAL TEST WORKFLOW

PHYSICALLY VALIDATED

▶ 1. Continuous Forward Movement Command Received



🚗 2. BUDDY Physically Traverses Test Track (4x Motors Active)



👋 3. Obstacle Placed in Path (< 0.25 m Detected by Ultrasonic)



🚫 4. Safety Supervisor Intervenes: Forces Immediate Wheel STOP



🔄 5. Obstacle Cleared: Safety Clear Signal → Forward Motion Resumes

1. TESTING HIERARCHY

Verification Levels

- **Unit Tests:** HAL, mock GPIO, safety calculations, config validator.
- **Hardware Tests:** PCA9685 I2C, 4x TT motors, dual TB6612, 4x HC-SR04.
- **Integration:** Sensor → ROS 2 → Supervisor → Motor Executor.
- **Regression:** Automated test suites on branch commits.

2. RESEARCH BENCHMARKING

Baselines & Metrics

- **Voice Intelligence:** Local vs. generic cloud STT (latency & accuracy) — *Benchmarking in progress.*
- **Vision / Emotion:** Edge inference latency & head tracking on Pi 4 — *Validation underway.*
- **Navigation / Safety:** Collision prevention rate & stop-resume reliability — *Physically validated.*

3. NFR COMPLIANCE

Engineering Qualities

- **Safety:** Fail-safe clamp & watchdog.
- **Privacy:** On-device data handling; zero unencrypted external streaming.
- **Reliability:** Hardware abstraction for zero-hardware simulation.
- **Maintainability:** Git workflow, structured logging & frozen interfaces.

4. RESPONSIBLE AI & PLSSE

Ethics & Disclosure

- **AI Tool Disclosure:** GenAI used for brainstorm, debugging & documentation; all code verified by team.
- **Ethics:** Emotion perception is probabilistic, not ground truth.
- **Honest Limitations:** Pi 4 compute constraints & lighting sensitivity acknowledged.

TARGET USERS & ADOPTION VALUE

Solo Living / Elderly

Accessible companionship, voice interaction, and indoor home check.

Smart-Home Enthusiasts

Privacy-preserving mobile robot with edge AI capabilities.

CURRENT PROTOTYPE HARDWARE COST ESTIMATE

LKR 80,000 – 100,000

**Estimated from current prototype hardware & electronics; final itemized BOM verification pending.*

DEFENSIBLE BOM ESTIMATE

Path to Productization: Custom integrated PCB (replaces loose wiring) → molded chassis → bulk component sourcing → local assembly support.

LIVE DEMONSTRATION ROADMAP

PP2 DEFENSE

- 1. **Voice Interaction:** Wake word & offline command
- 2. **Vision Perception:** Face detection & active neck tracking
- 3. **Physical Locomotion:** Autonomous patrol execution
- 4. **Obstacle Intervention:** Ultrasonic detection & instant STOP
- 5. **Safe Recovery:** Path clearance & patrol resumption
- 6. **Application Layer:** Telemetry & state monitoring

"The demonstration proves one integrated system, not isolated components."

BUDDY Physical Component Bill

Subsystem	Components Used	Interface
Main Compute	Raspberry Pi 4 Model B (4GB)	Ubuntu / ROS 2 Core
Motor Control	2x TB6612FNG Dual Drivers + 4x TT Motors	GPIO (PWM / Direction)
Pan/Tilt Neck	PCA9685 16-Ch PWM + 2x MG90S Servos	I2C Bus (0x40)
Sensors & Level	4x HC-SR04 + 4-Ch BSS138 Converter	5V Echo to 3.3V Logic
Vision & Audio	IMX219 CSI Camera + 2x AB13X Mics + Speaker	CSI / USB Audio Hub

Dual-Rail Power Isolation Architecture

RAIL A (Motor & Servo High Current):
3x 18650 Li-ion Cells (11.1V – 12.6V) → 3S BMS → In-Line Fuse → Main Power Switch → LM2596S Buck (Step down to 6.0V Regulated) → TB6612FNG VM & PCA9685 V+

RAIL B (Isolated Logic Compute):
Dedicated 5V / 3A USB-C Power Bank → Raspberry Pi 4 Model B (Zero back-EMF interference from inductive DC motors).

Grounding & Protection: Shared common ground plane across all subsystems; BSS138 bidirectional level shifters strictly prevent 5V ultrasonic Echo signals from over-volting 3.3V Raspberry Pi GPIO pins.

Safety State Machine Execution Cycle

STATE: CLEAR (All distances > 0.45 m) NORMAL EXECUTION

↓ Obstacle detected at ≤ 0.45 m

STATE: CAUTION (0.25 m < Dist ≤ 0.45 m) SPEED REDUCTION 50%

↓ Obstacle penetrates safety envelope ≤ 0.25 m

STATE: EMERGENCY STOP (Dist ≤ 0.25 m) ZERO MOTOR PWM

↑ Sensor clears for consecutive debounce frames (> 0.35 m hysteresis)

STATE: RECOVERY / RESUME CONTROLLED RESTART

Edge Case Handlers

- **Sensor Loss Timeout:** If ultrasonic echo fails for > 250ms, auto-stop triggers.
- **Hysteresis Recovery:** Prevents rapid stop-start oscillation at boundary limits.
- **Watchdog Heartbeat:** Loss of ROS 2 movement publisher triggers fail-safe idle.

Servo Bounds Clamping

- **Pan (Horizontal):** Hardware clamped to 30° – 150°.
- **Tilt (Vertical):** Hardware clamped to 45° – 120°.
- Prevents physical collision between camera housing and chassis.

Auditory Intelligence

BENCHMARKING

Baseline Comparison:

Cloud-dependent generic ASR pipeline vs. BUDDY local edge-optimized model.

Evaluation Metrics:

- Wake-word trigger rate & false positive rate
- Command intent recognition accuracy
- Offline inference response latency (ms)
- Performance under ambient room noise

**Current Status: Voice pipeline integrated on device; quantitative benchmarking underway.*

Vision & Emotion

VALIDATION

Baseline Comparison:

Static desktop CNN classifier vs. Active robotic pan/tilt tracking pipeline.

Evaluation Metrics:

- Face detection & ID classification accuracy
- Inferred emotion class confidence
- On-device FPS & inference compute latency
- Servo tracking responsiveness under movement

**Current Status: Hardware pipeline running on Pi 4; frame-rate optimization in progress.*

Navigation & Safety

PHYSICALLY TESTED

Baseline Comparison:

Open-loop motor drive vs. Safety-supervised real-time obstacle interception.

Evaluation Metrics:

- Collision avoidance success rate (100% on tested conditions)
- Stop reaction time from ultrasonic trigger
- Autonomous patrol loop completion consistency
- Path clearance recovery reliability

***Current Status: Physical obstacle-triggered stop-and-resume validated on prototype.**

Prototype Hardware Estimate Categories

LKR 80,000 – 100,000

Subsystem Category	Key Elements Included	Status
Embedded Compute	Raspberry Pi 4 Model B (4GB), MicroSD, USB Hub	Verified Prototype
Perception Stack	IMX219 CSI Camera, Dual USB Mics, 4x HC-SR04	Verified Prototype
Actuation & Drivers	4x TT Motors, 2x TB6612, PCA9685, 2x MG90S	Verified Prototype
Power Management	Samsung 18650 3S, 3S BMS, Buck, USB-C Bank	Verified Prototype
Chassis & Harness	Acrylic chassis, Level shifters, Wiring, Switches	Verified Prototype

**Note: Presented as an engineering hardware estimate based on single-unit component sourcing; final audited BOM verification pending.*

Engineering Path to Commercial Viability

1. Custom Carrier PCB: Integrates TB6612, PCA9685, level converters, and buck regulators into one board—eliminating 80% of loose point-to-point wiring.

2. Molded Modular Chassis: Replace acrylic assembly with lightweight injection-molded structure with integrated sensor mounts.

3. Volume Component Sourcing: Transition from off-the-shelf single breakout boards to bulk BOM sourcing, drastically reducing per-unit cost.

4. Local Assembly & Support: Enables low-cost domestic deployment, accessible spare parts, and modular software updates.

Module / Subsystem	Primary Lead	Implementation Evidence	Status Label
ROS 2 Architecture & Core HAL	Team Integration	Frozen msgs, centralized logging, YAML config, mock GPIO backend	VERIFIED
Safety Supervisor Node	SSY / Team	Obstacle interception (0.45m/0.25m), watchdog timer, servo limits	VERIFIED
4-Wheel Motor Actuation	SSY Wickramasinghe	Dual TB6612FNG drivers, TT geared motors, speed/direction executor	VERIFIED
4-Corner Ultrasonic Sensing	SSY Wickramasinghe	HC-SR04 array with BSS138 level conversion & debounced ROS publisher	VERIFIED
Pan/Tilt Head Mechanism	Sobiya Anton Suresh	PCA9685 I2C driver, dual MG90S servos, safe angular bounds	VERIFIED
Offline Voice Intelligence	Ashandth U.	Local wake-word engine & offline speech intent parsing pipeline	BENCHMARKING
Face & Emotion Vision	Sobiya Anton Suresh	IMX219 CSI camera capture, face detection & emotion classification	VALIDATION
Web/Mobile Interaction Layer	SSY / Team	Status monitoring interface & manual override controls where implemented	INTEGRATED

PROJECT ID: R26-SE-040 • FINAL-YEAR RESEARCH PP2

Ready for Live Demonstration & Q&A

BUDDY demonstrates how an affordable, edge-oriented platform combines natural human interaction, autonomous home monitoring, and disciplined safety-first engineering.

ASHANDTH UTHAYASHANKAR

Offline Auditory Intelligence

SOBIYA ANTON SURESH

Face & Emotion Perception

SSY WICKRAMASINGHE

Navigation, Safety & Web Layer

SLIIT Faculty of Computing — Software Engineering Specialization

Image Sources



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